MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1108.4
STANDARD **Interpretability and Quality Local Set**	**02 March 2023**

1 Scope

This document defines metadata for documenting image metrics and compression characteristics within a Motion Imagery stream or file. Such metadata facilitates documenting the quality of Motion Imagery along the imaging workflow with objective metrics or subjective interpretability criteria providing valuable information to the end user. Encoding the metadata as binary KLV (Key Length Value) supports encapsulation in MISP-approved containers. Metric computation is location based defined by a starting Frame and an ending Frame.

2 References

[1] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.
[2] MISB ST 0807.26 MISB KLV Metadata Registry, Jun 2022.
[3] MISB ST 0603.5 MISP Time System and Timestamps, Oct 2017.
[4] IEEE 754-2008 Standard for Floating-Point Arithmetic [and Floating-Point formats].
[5] MISB ST 0901.3 Video-National Imagery Interpretability Rating Scale, Oct 2017.
[6] BIPM The International System of Units (SI) 9th Edition, May 2019.
[7] MISB MISP-2023.2 Motion Imagery Standards Profile, Mar 2023.
[8] MISB MISP-2023.2: Motion Imagery Handbook, Mar 2023.

3 Revision History

Revision	Date	Summary of Changes
ST 1108.4	03/02/2023	• Added “Metric Units” Item 7 to Metric Local Set • Added Req -17 • Bounded Metric Local Set lengths in Table 5 • Added default units to Table 6 • Changed “Compression Ratio” item to Optional (O) • Revised Figure 3 to include Metric Units • Removed UL (keys) from items listed • Changed “element” to “item” for KLV items consistent with current MISB practice • Changed “Deprecated” to “Retired” Appendix A requirements • Added acronyms

		• Updated references

4 Acronyms and Definitions

CRC	Cyclic redundancy check
DLP	Defined Length Pack
GSD	Ground Sample Density
KLV	Key-Length-Value
MSSIM	Multi-scale structural similarity index measure
PSNR	Peak signal-to-noise ratio
SI	International System of Units
SNR	Signal-to-noise ratio
SSIM	Structural similarity index measure
ST	Standard
VNIIRS	Video National Imagery Interpretability Rating Scale
Interpretability	The potential of an analyst to extract specific intelligence data for an intelligence task
Quality	Measures the overall appearance of Motion Imagery

5 Introduction

MISB ST 1108 defines the content and structure for Motion Imagery Interpretability and Quality metadata that documents imagery metrics and image compression parameters for a Motion Imagery stream or file. The intent and principles for this standard include:

- Agnostic to metric type
    - Support for VNIIRS predictive metric and other image-based metrics
- Document location in data lifecycle of metric calculation
    - Ability to identify various forms of the same algorithm calculated at the same location over time
- Document Motion Imagery stream characteristics at time of metric calculation
    - Motion Imagery timestamps and encoding information for metric comparisons during data lifecycle

6 Interpretability and Quality Local Set

MISB ST 0107 [1] provides a set of baseline requirements for using KLV in Motion Imagery.

Requirement	
ST 1108.2-01	All metadata shall be in accordance with MISB ST 0107.

The Interpretability and Quality Local Set (LS) 16-Byte Universal Label (key) registered in MISB ST 0807 [2] is:

06.0E.2B.34.02.03.01.01.0E.01.03.03.1C.00.00.00 (CRC 6633)

Table 1 summarizes the Interpretability and Quality LS with the columns defined as:

- **Tag** indicates the KLV LS tag number for the item; the tag is a one-byte integer encoded as a BER-OID value
- **Item Name** is the name of the dictionary item
- **Type** specifies the data type when encoding the value into binary; generalizing the Types uint and int allow the application to define the number of bytes needed for a values representation
- **Length** is the maximum length of the item's value; lengths denoted as "V" are variable and computed at runtime' per ST 0107, use the fewest number of bytes for efficiency
- **M/O** indicates if the LS item is Mandatory (M) or Optional (O)
- **Notes** provide additional details about the value

Table 1: Interpretability and Quality Local Set

Tag	Item Name	Type	Length	M/O	Notes
1	Assessment Point	enum	1	M	Location in workflow metric evaluated specified as an enum (see Table 2)
2	Metric Period Pack	dlp	12	M	Period within metric calculated (see Table 3)
3	Window Corners Pack	dlp	V	O	Image sub-region upper left, lower right row/column positions (see Table 4). Full image assumed when not present.
4	Metric LS	Set	V	M	Metric Name; Metric Version; Metric Parameters, Metric Time, and Metric Value (see Table 5)
5	Compression Type	enum	1	M	0 = uncompressed 1 = H.262 2 = H.264 3 = H.265 4 = JPEG2000 5-255 Reserved
6	Compression Profile	enum	1	M	0 = N/A (uncompressed) 1 = Main (H.264) 2 = Main 10 (H.265) 3 = Constrained Baseline (H.264) 4 = High (H.264) 5 = Main 4:2:2 12 (H.265) 6 = Main 4:4:4 12 (H.265) 7 = High 4:2:2 (H.264) 8 = High 4:4:4 Predictive (H.264) 9-255 = Reserved
7	Compression Level	utf8	3	M	"N/A" for uncompressed Expressed as a String; ex: "5", "4.2", "5.1"

8	Compression Ratio	float	8	O	Source-to-compressed size; ex: 25.2 means 25.2:1. Set to 1.0 for uncompressed
9	Stream Bitrate	uint	2	M	Expressed in kilobits/sec
10	Document Version	uint	1	M	Version number of this standard
11	CRC-16-CCITT	uint	2	M	Two-byte check value on LS

6.1 Assessment Point

An Assessment Point is a physical location along the imaging workflow. The Assessment Point item documents the calculation of one or more metrics at this location. The Assessment Point is an enum; Table 2 enumerates the allowed values for Assessment Point. This LS item is mandatory.

Requirement	
ST 1108.3-06	An Assessment Point item shall be present in an Interpretability and Quality Local Set.

Table 2: Assessment Point Identifier

Assessment Point	Description	Value
Sensor	Output directly from sensor	1
Sensor Encoder	Output from Motion Imagery encoder (if applicable)	2
GCS Received	Platform to ground station reception point	3
GCS Transmit	GCS transmission point	4
Library/Archive	Receipt at archival location	5

6.2 Metric Period Pack

The Metric Period Pack documents the period over which a calculated metric applies indicated by a beginning image Frame timestamp (i.e., Start Time) and an ending period image Frame time offset (i.e., Time Offset). This Local Set item is mandatory and encoded as a defined length pack (DLP) (see Figure 1) where the beginning Frame timestamp is a Precision Time Stamp as specified in MISB ST 0603 [3] and the ending Frame timestamp calculated as a positive offset added to the beginning Frame timestamp. Thus, the difference between the two numbers is the period defined for the metric calculation.

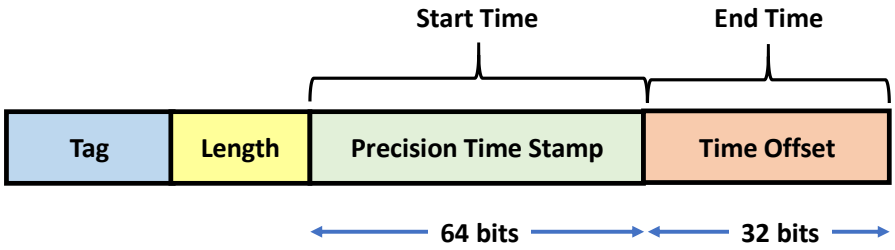


Figure 1: Metric Period Pack

Requirement	
ST 1108.3-07	A Metric Period Pack shall be present in an Interpretability and Quality Local Set.

Table 3 defines the items of the Metric Period Pack. The Element Name is the identifier associated with the value. The Data Type specifies the bit representation of the value. The Units specify the units of the value. The Notes provide additional information about the value.

Table 3: Metric Period Pack Elements

Metric Period Pack				
Element Name	**Data Type**	**Length (Bytes)**	**Units**	**Notes**
Time Start	uint	8	µs	Timestamp marking the beginning of the metric calculation
Time Offset	uint	4	µs	Timestamp offset from Precision Time Stamp marking the end of the metric calculation

6.3 *Window Corners Pack*

The Window Corners Pack, which is a defined length pack (DLP), describes a sub-image region within an image Frame for the metric calculation bounded by Frame Line[start] to Frame Line [end] and Frame Row[start] to Frame Row[end]. Figure 2 illustrates a window extracted as a sub-region within an image. C_1 denotes Frame Line[start] and Frame Row[start], while C_2 denotes Frame Line[end] and Frame Row[end]. This enables defining a local region within an image to calculate image metrics rather than over the entire image.

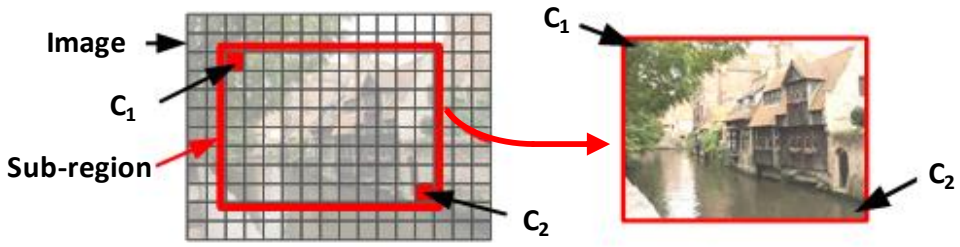


Figure 2: Window as sub-region of an image

Table 4 defines the Window Corners Pack used to encode the corner information of a sub-region or "window" within an image Frame. The Element Name is the identifier for the value. The Type specifies the bit representation of the value; values start at zero. Length is the number of bytes used for the value. Since all values in the Window Corners Pack are BER, the length is variable and is determined during pack construction or parsing. The Units specify the units of the value. In the Window Corners Pack, the units are coordinates in the image Frame, so each value is in units of Pixels. All four values of rows or columns in the Window Corners Pack are zero based, i.e., each value starts at zero. The Window Corners Pack item is optional.

Table 4: Window Corners Pack Elements

Window Corners Pack			
Element Name	**Type**	**Units**	**Notes**
C_1 Row	BER-OID	Pixels	Row for the upper left corner of the window. First row starts at zero.
C_1 Column	BER-OID	Pixels	Column for the upper left corner of the window. First column starts at zero.
C_2 Row	BER-OID	Pixels	Row for the lower right corner of the window. First row starts at zero.
C_2 Column	BER-OID	Pixels	Column for the lower right corner of the window. First column starts at zero.

In the absence of the Window Corners Pack, the metrics are for the whole image frame.

Requirement	
ST 1108.3-08	Where a metric represents a sub-region of an image Frame, the Window Corners Pack shall be present in an Interpretability and Quality Local Set.

6.4 Metric Local Set

Table 5 summarizes the Metric LS with the columns defined as:

- **Tag** indicates the KLV LS tag number for the item; the tag is a one-byte integer encoded as a BER-OID value
- **Item Name** is the name of the dictionary item
- **Type** specifies the data type when encoding the value into binary. Note: generalizing the Types uint and int allows the application to define the number of bytes needed for a values representation.
- **Length** is the maximum length of the item's value; per ST 0107, use the fewest number of bytes for efficiency
- **M/O** indicates if the LS item is Mandatory (M) or Optional (O)
- **Notes** provide additional details about the value

Table 5: Metric Local Set Items

Metric Local Set					
Tag	**Item Name**	**Type**	**Length**	**M/O**	**Notes**
1	Metric Name	utf8	128	M	Examples “VNIIRS;” “RER” (Relative Edge Response)
2	Metric Version	utf8	32	M	Alphanumeric denoting calculated values: “Human” for observed
3	Metric Implementer	utf8	128	M	Identifies the organization responsible for the calculation of the metric
4	Metric Parameters	utf8	128	O	Additional information necessary for repeatability
5	Metric Time	uint	8	M	Time of metric assessment (defined with MISP Precision Time Stamp)
6	Metric Value	float	8	M	Numeric value of calculation (IEEE format [4])
7	Metric Units	utf8	128	O	Units of metric in SI

This standard requires at least one instance of the Metric LS per Interpretability and Quality LS; however, the Metric LS may be used multiple times. This enables defining multiple metrics within the timeframe defined by the Metric Period Pack.

Requirement	
ST 1108.3-09	At least one (1) Metric Local Set shall be present in an Interpretability and Quality Local Set.

Figure 3 shows an example of an Interpretability and Quality LS with two metrics “VNIIRS” and “GSD” included as two instances of Tag 4 – Metric LS. Note: mandatory items only shown.

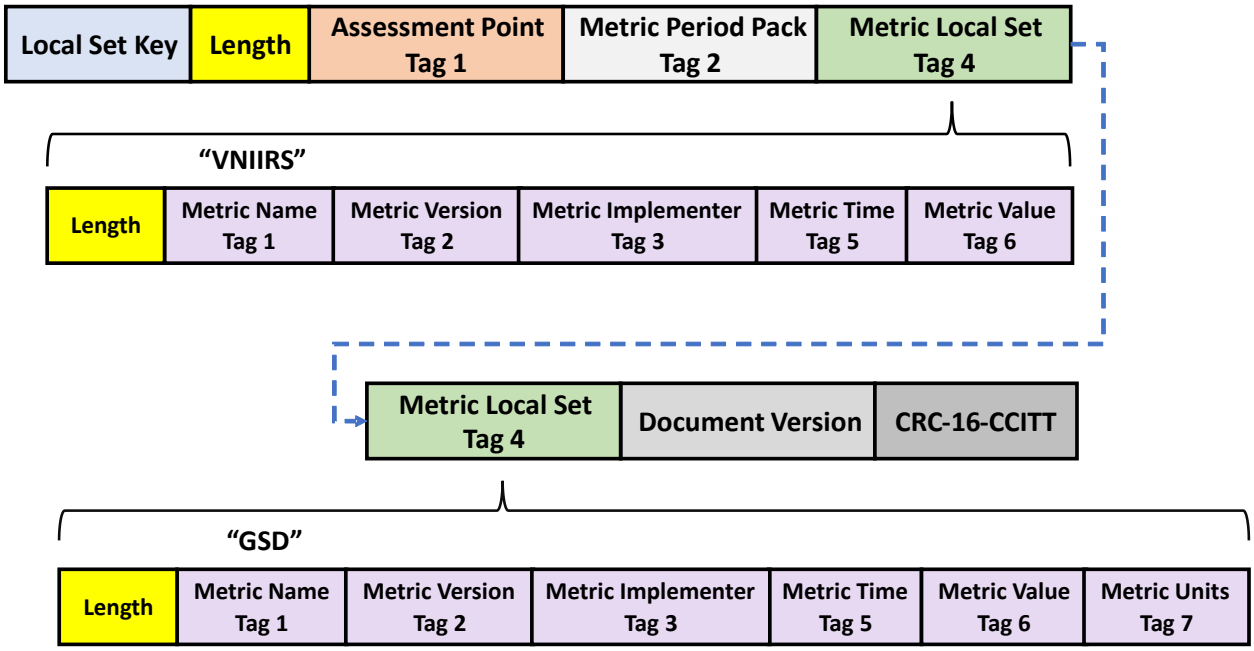


Figure 3: Example: repeated use of Metric LS tag for VNIIRS and GSD metrics.

6.4.1 Metric Name

The Metric Name is the name of the metric calculated. Although its type is a string MISB defines the following strings for consistency. (see Table 6). Table 6 also specifies default units for metrics; those designated as "(unitless)" are metrics without units. The Metric Units (Item 7) provides for overriding the default unit designations and other metrics not indicated here per Section 6.4.7.

Table 6: Metric Assigned Names

Metric Name	Meaning	Default Units
GSD	Ground Sample Distance	m
MSSIM	Multi-Scale Structural Similarity Index	(unitless)
PSNR	Peak Signal-to-Noise Ratio	dB
RER	Relative Edge Response	(unitless)
SNR	Signal-to-Noise Ratio	dB
SSIM	Structural Similarity Index	(unitless)
VNIIRS	Video-National Imagery Interpretability Rating Scale (MISB ST 0901 [5])	(unitless)

Users may elect metrics other than those listed in Table 6; however, the string cannot exceed the maximum length for the item as given in Table 5. The MISB strongly encourages using the Units (Item 7) to avoid data misinterpretation.

6.4.2 Metric Version

The Metric Version item serves a dual role. Firstly, it indicates how the method is used to calculate the metric, and secondly, it provides configuration management information. When generated by a human observer, the value is "human." When generated by an algorithm, the value is an alphanumeric configuration management identifier for the algorithm.

Requirement(s)	
ST 1108.3-10	Where the Metric Version is produced by a human, the value of the Metric Version shall be "human."
ST 1108.3-11	Where the Metric Version is produced by an algorithm, the value of the Metric Version shall be an alphanumeric composed of characters and numbers.

6.4.3 Metric Implementer

The Metric Implementer item identifies the organization responsible for the method used to calculate the metric. This enables various versions of the same metric without having to maintain an active registry. The string should be Name of Company/Organization[ASCII Code 30]Subgroup. ASCII Code 30 is Record Separator and used here to separate two meaningful strings with a default separator. For example: "Company XYZ[ASCII Code 30]Imaging Dept."

Requirement	
ST 1108.3-12	The format for the value of the Metric Implementer shall be Name of Company/Organization[ASCII Code 30]Subgroup, where ASCII Code 30 designates Record Separator.

6.4.4 Metric Parameters

Metric Parameters item is a user-defined string for specifying any unique conditions used in the computation. For example, within the period of calculation all Frames used and averaged, are several Frames skipped, etc. This information ensures repeatable calculations, or the same process applied downstream for like-metric comparisons.

6.4.5 Metric Time

Metric Time item is the time of metric assessment, which with respect to the Metric Time pack may be a future value. Metric Time is a microsecond Precision Time Stamp as defined in MISB ST 0603.

6.4.6 Metric Value

The Metric Value item is a floating-point numeric based on the resultant calculation. The length field of the Metric Value KLV item dictates the length of the corresponding Metric Value.

6.4.7 Metric Units

The Metric Units item provides the units for the measured metric. Units are SI (International System of Units) [6]. Table 6 lists the default units for metrics in this document.

Requirement	
ST 1108.4-17	All metric units shall be in SI (International System of Units).

6.5 Compression Type

The Compression Type item is a MISP-approved [7] type such as H.264, H.265 etc. specified using an unsigned 8-bit integer enumeration indicated in Table 1. For no compression (i.e., uncompressed) specify a value of zero (0).

6.6 Compression Profile

The Compression Profile item is a valid MISP-approved level for MPEG-2, H.264 and H.265 specified using an unsigned 8-bit integer enumeration value indicated in Table 1. For no compression (i.e., uncompressed) specify a value of zero (0).

6.7 Compression Level

The Compression Level item is a valid MISP-approved level for H.262, H.264 and H.265 specified as a string value. Example typical values for a compression level are 4, 4.1, 4.2, 5 for H.264/H.265 and ML, HL for H.262. Thus, the string length for this item is three characters or less. See the MISP for allowed levels. For no compression (i.e., uncompressed) specify a value of "N/A."

6.8 Compression Ratio

The Compression Ratio item, expressed as an IEEE floating point value, is the ratio of source to compressed signal data quantity. In cases where the source characteristics are not known and the compression ratio not computable do not include this item in the LS. For no compression (i.e., uncompressed) specify a value of 1.0.

6.9 Stream Bitrate

The Stream Bitrate item is the measured or known bitrate of the stream at the metric calculation time. Stream Bitrate is a 16-bit unsigned integer measuring kilobits per second. For example, 2200 Kbits/sec represents a stream bit rate of 2.2 Mbits/sec.

6.10 Document Version

The Document Version item indicates the numeric version of this document. For example, ST 1108.3 is version 3, ST 1108.4 is version 4, etc.

Requirement	
ST 1108.3-13	A Document Version item shall be present in an Interpretability and Quality Local Set.

6.11 CRC-16-CCITT

To detect erroneous metadata post transmission, the Local Set includes a two-byte CRC as the last item computed across the entire LS starting with the 16-byte LS key and ending with the length field of the CRC data item. If the calculated CRC of the received LS packet does not match the CRC stored in the packet, the packet is invalid and discarded.

The Motion Imagery Handbook [8] Appendix D provides the algorithm for calculating the CRC.

Requirement(s)	
ST 1108.3-14	The Interpretability and Quality Local Set shall contain a CRC-16-CCITT.
ST 1108.3-15	The calculation of the CRC-16-CCITT shall be in accordance with the MISB Motion Imagery Handbook.
ST 1108.3-16	The CRC-16-CCITT shall be the last item in the Interpretability and Quality Local Set.

7 Appendix A – Retired Requirements

Requirement(s)	
ST 1108.2-02 (Retired)	The Interpretability and Quality (IQ) metadata set shall use Local Data Set 16-byte Universal Key 06.0E.2B.34.02.03.01.01.0E.01.03.03.1C.00.00.00 (CRC 6633) for its implementation.
ST 1108.2-03 (Retired)	Chip sizes shall be 32, 64 or 128.
ST 1108.2-04 (Retired)	For a given image Frame only one chip shall be allowed.
ST 1108.2-05 (Retired)	The Chip Frequency Ratio shall have a cut-off frequency of 0.15.